Project Grains Sales Data Visualization Using Python, Power BI and Tableau

AIM

To analyze the Grains Sales dataset and create three visualizations - bar chart, line chart, and pie chart - using Python, Power BI, and Tableau to understand sales performance, identify top-selling grains, and analyze sales distribution across categories.

PROBLEM STATEMENT

Grain traders generate large amounts of sales data through daily orders. Understanding this data manually is difficult because it contains order dates, grain types, categories, regions, quantities, and prices. This project uses Python libraries such as Pandas and Matplotlib, along with Power BI and Tableau, to process the dataset and create visualizations. These visualizations help identify high-performing grains, observe monthly sales trends, and understand category-wise sales contributions.

SOFTWARE REQUIREMENTS

- Platform: Google Colab, Power BI Desktop, Tableau Desktop
- Programming Language: Python
- Dataset Format: CSV (.csv)
- Libraries: Pandas, Matplotlib
- Input Dataset: Grains Sales CSV file (Grains_Dataset.csv)

DATASET DESCRIPTION

The Grains Sales dataset contains 600 order records with the following columns.

Column Name	Description
Order_ID	Unique ID of the order
Order_Date	Date of the order
Grain	Grain sold (Rice, Wheat, Maize, Ragi, Bajra, Jowar, Toor Dal, Moong Dal, Chana Dal, Urad Dal)
Category	Category of the grain (Cereal, Millet, Pulse)
Region	Region of sale (Chennai, Coimbatore, Madurai, Salem, Trichy)
Quantity_kg	Quantity sold in kilograms
Price_per_kg	Price per kilogram
Sales	Sales amount of the order

1. PYTHON

PROCEDURE

1. Import the dataset into Python, Power BI, and Tableau.
2. Inspect the dataset and identify the date, grain, category, quantity, and price columns.

3. Handle missing values and ensure that dates and numerical columns have the correct data types.
4. Calculate total sales by multiplying Quantity_kg by Price_per_kg.
5. Create a bar chart to compare values across categories.
6. Create a line chart to analyze trends.
7. Create a pie chart to show the distribution across categories.
8. Use Pandas and Matplotlib to generate the three visualizations in Python.
9. Import the dataset into Power BI and Tableau and recreate the same visualizations.
10. Compare the charts and identify trends, popular categories, and business insights.

Program and Output

```
import pandas as pd
import matplotlib.pyplot as plt

# Read the dataset
df = pd.read_csv("Grains_Dataset.csv")

# Clean column names
df.columns = df.columns.str.strip()

# Convert required columns
df["Order_Date"] = pd.to_datetime(df["Order_Date"], errors="coerce")
df["Quantity_kg"] = pd.to_numeric(df["Quantity_kg"], errors="coerce")
df["Price_per_kg"] = pd.to_numeric(df["Price_per_kg"], errors="coerce")

# Remove missing values
df = df.dropna(subset=["Order_Date", "Grain", "Category",
                       "Quantity_kg", "Price_per_kg"])

# Calculate total sales
df["Total Sales"] = df["Quantity_kg"] * df["Price_per_kg"]

# 1. BAR CHART - Sales by Grain
grain_sales = df.groupby("Grain")["Total Sales"].sum().sort_values(ascending=False)

plt.figure(figsize=(8, 5))
grain_sales.plot(kind="bar", color="goldenrod")
plt.title("Sales by Grain")
plt.xlabel("Grain")
plt.ylabel("Total Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

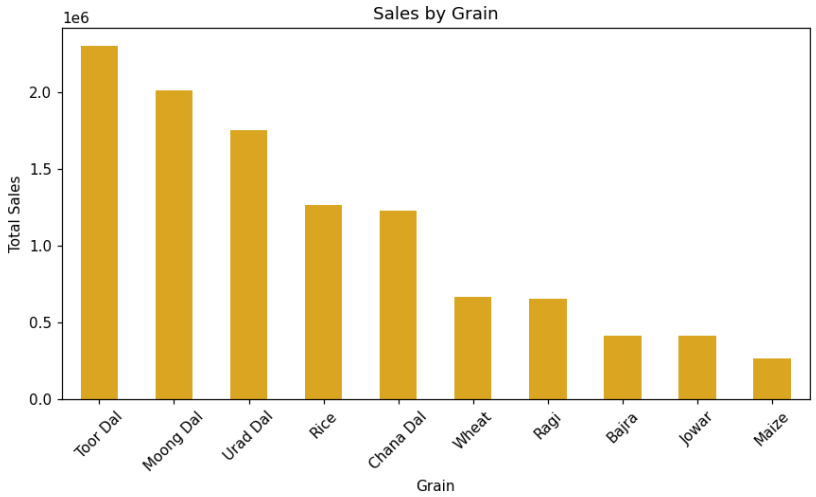


Fig 1: Bar chart - Sales by Grain

```
# 2. LINE CHART - Monthly Sales Trend
monthly_sales = df.groupby(df["Order_Date"].dt.to_period("M"))["Total Sales"].sum()

plt.figure(figsize=(9, 5))
plt.plot(monthly_sales.index.astype(str), monthly_sales.values,
         marker="o", color="darkgreen")
plt.title("Monthly Grain Sales Trend")
plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.xticks(rotation=45)
plt.grid(True)
plt.tight_layout()
plt.show()
```

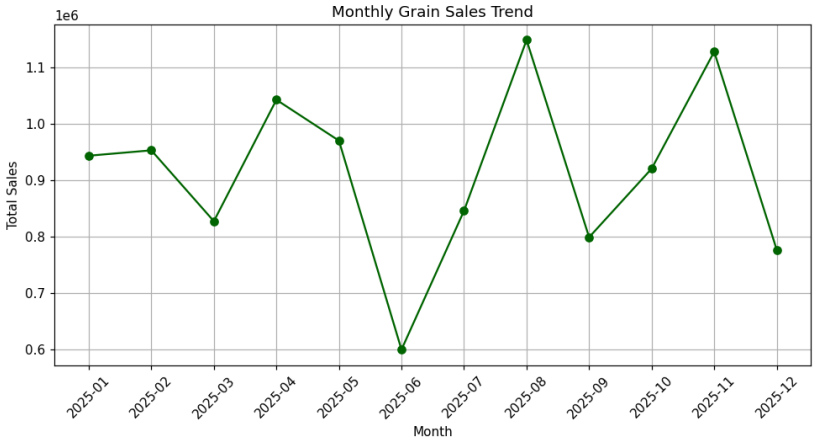


Fig 2: Line chart - Monthly Sales Trend

```
# 3. PIE CHART - Category-wise Sales Distribution
category_sales = df.groupby("Category")["Total Sales"].sum()

plt.figure(figsize=(6, 6))
plt.pie(
    category_sales.values,
    labels=category_sales.index,
    autopct="%1.1f%%",
    startangle=90
)
plt.title("Category-wise Sales Distribution")
plt.tight_layout()
plt.show()
```

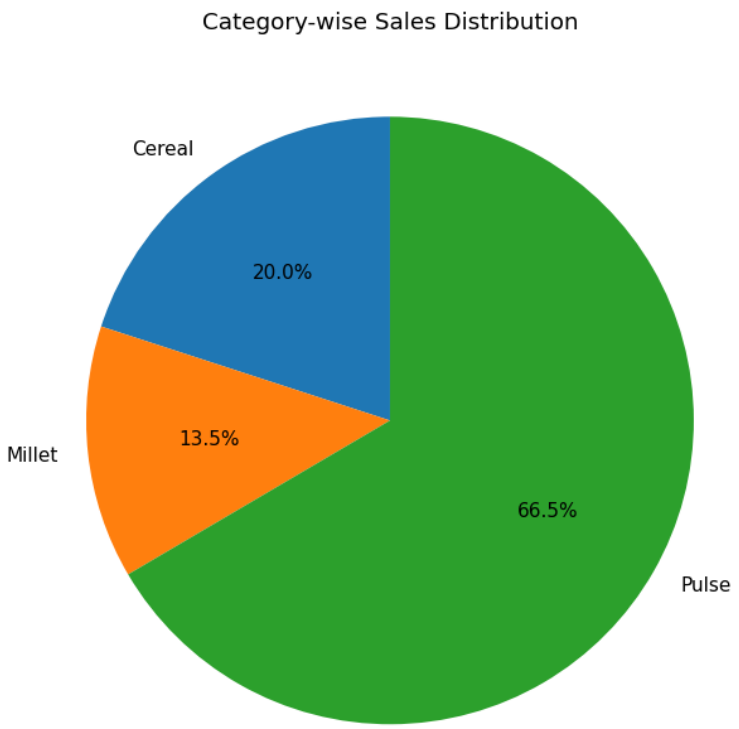


Fig 3: Pie chart - Category-wise Sales Distribution

2. POWER BI

1. Open Power BI Desktop and select Home > Get Data > Text/CSV.
2. Select the Grains_Dataset.csv file and click Load.
3. Check the data types of Order_Date, Quantity_kg, Price_per_kg, and Category.
4. Create a calculated column using Table tools > New column.
   Total Sales = 'Grains'[Quantity_kg] * 'Grains'[Price_per_kg]
5. Create a bar chart by placing Grain on the Y-axis and Total Sales on the X-axis, summarized by Sum.
6. Create a line chart using Order_Date (Month) on the X-axis and Sum of Total Sales on the Y-axis.
7. Create a pie chart using Category in Legend and Sum of Total Sales in Values.
8. Add Card visuals to display Total Sales, Total Quantity, and Total Orders.
9. Add a Region or Category slicer to filter the report.
10. Arrange the visuals, format titles and colors, and save the report as Grains_Sales.pbix.

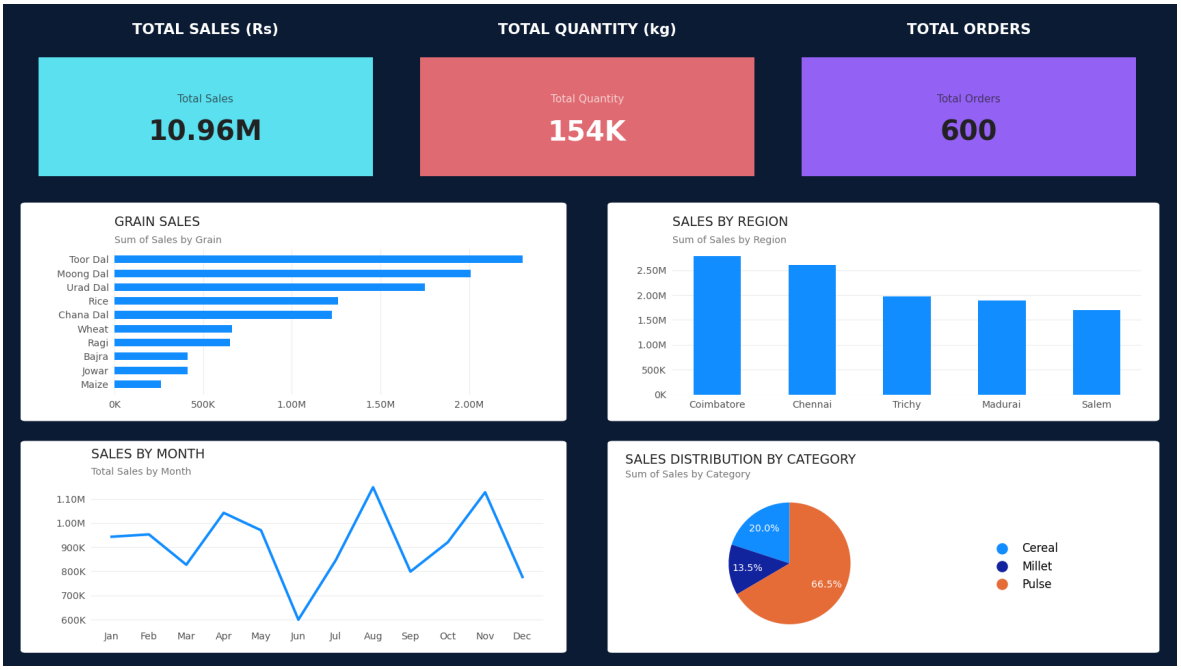


Fig: Grains Sales Dashboard (Power BI)

3. TABLEAU

1. Open Tableau Desktop.
2. Under Connect, select Text File and open Grains_Dataset.csv.
3. Open a new worksheet.
4. Create the bar chart by placing Grain on Rows and Sales on Columns, then sort in descending order.
5. For the line chart, use Order_Date (Month) on Columns and Sales on Rows, then select Line in the Marks card.
6. Create a pie chart by selecting Pie in the Marks card, placing Category on Color and Sales on Angle.
7. Set the aggregation to Sum where appropriate.
8. Add chart titles and customize the colors.
9. Select New Dashboard and drag all the worksheets onto the dashboard.
10. Save the workbook as Grains_Sales.twbx or .twb.

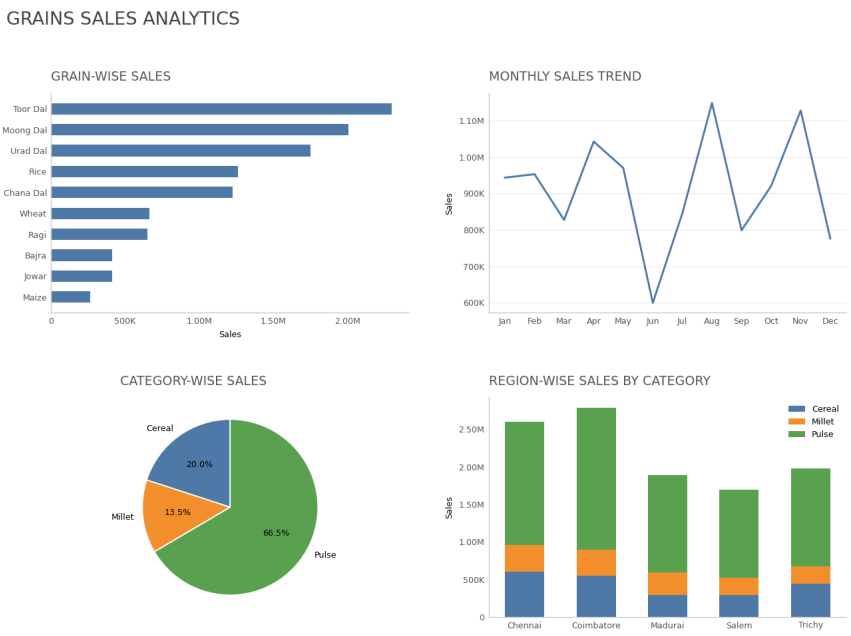


Fig: Grains Sales Dashboard (Tableau)

VISUALIZATIONS

1. Bar Chart - Sales by Grain

Compares total sales of each grain to identify the best-selling grain.

2. Line Chart - Monthly Sales Trend

Shows how total sales change from month to month during the year.

3. Pie Chart - Category-wise Sales Distribution

Displays the share of Cereal, Millet, and Pulse categories in total sales.

OBSERVATION AND INSIGHTS

- Toor Dal has the highest sales, followed by Moong Dal and Urad Dal. The lowest-selling grain is Maize.
- Pulses contribute 66.5% of total sales, Cereals 20.0%, and Millets 13.5%.
- Monthly sales are highest in August and lowest in June, showing seasonal variation.
- Coimbatore is the top-selling region and Salem the lowest.
- Python, Power BI, and Tableau provide different ways to create and present the same visual analysis.

RESULT

The Grains Sales dataset was used to create bar charts, line charts, and pie charts using Python, Power BI, and Tableau. The visualizations support comparison of grain-wise sales, analysis of monthly trends, and category-wise sales distribution using three different data visualization tools.